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$U, V \sim$ iid $N(0,1)$

$f_{UV}(u,v) = \frac{1}{2\pi} e^{-\frac{(u^2+v^2)}{2}}$, $u, v \in \mathbb{R}$

$\mu_U = \mu_V = 0 \quad \sigma_U = \sigma_V = 1$

$\rho_{UV} = 0$

$\rho \in [-1, 1]$

$X = U$

$Y = \rho U + \sqrt{1-\rho^2}\, V$

$\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)} = \frac{1}{2}\left(x^2 + \frac{(y-\rho x)^2}{1-\rho^2}\right)$

$f_{XY}(x,y) = \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)}}$, $x, y \in \mathbb{R}$

$\mu_X = \mu_Y = 0 \quad \sigma_X = \sigma_Y = 1$

$\rho_{XY} = \rho$

$\tilde{X} = \sigma_1 X + \mu_1$

$\tilde{Y} = \sigma_2 Y + \mu_2$

$\frac{\tilde{X} - \mu_1}{\sigma_1} = U$

$\frac{\tilde{Y} - \mu_2}{\sigma_2} = \rho U + \sqrt{1-\rho^2}\, V$

$f_{\tilde{X}\tilde{Y}}(\tilde{x}, \tilde{y}) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{\tilde{x}, \tilde{y}}{2(1-\rho^2)}}$

$\Pr(2\tilde{X} + 3\tilde{Y} < 4) = \Phi_Z\left(\frac{4 - 2\mu_1 - 3\mu_2}{\sqrt{4\sigma_1^2 + 9\sigma_2^2}}\right)$

W

PVS

$X \in \mathbb{R}$

extended real line $+\infty, -\infty$

$\Pr(X = +\infty) > 0$

$U \in \{1, 2\}$ $\quad \Pr(U=1) = \frac{1}{2}$, $\Pr(U=2) = \frac{1}{2}$

$X = \frac{1}{|1-U|}$

$\Pr(X = \infty) = \frac{1}{2}$

$\Pr(X \text{ finite}) = \frac{1}{2}$

Cauchy distribution

"finite w.p. 1"

$\lim_{n \to \infty} P(X > n) = 0$